

COSMOS Docket

US provisional packets for Legal

Not a USPTO filing. Not legal advice. Not a novelty opinion. Keith + Legal file. This TUI does not click USPTO.

September 2026 · micro-entity \$65 per provisional

Umbrella: the new way - reliable, scalable AI. Locally deployable, low fixed cost, free-weight models. Porosity: different-mistake overlay covers faster than copies.

MOTIF 1 - Docket (patents / trademarks / IP)

Keith 2026-09-07. Move most of the build private; leave provenance that COSMOS has been running since early July as BTS-MESH; file provisionals on COSMOS/MOTIF and whatever else qualifies. Shoot for ~10-12 if they are \$65 each. Umbrella: *the new way - reliable, scalable AI*. Locally deployable, low fixed cost, free-weight models. **Porosity** (Keith: Pouroosity): coder quality as hole-set; low-quality models with **different mistakes**, overlaid, cover surface quickly at a > **linear** rate versus stacking copies.

This file is the index. Written-description packets (Legal, not USPTO) live in docs/research/docket/. Vetting: docs/research/docket/VETTING.md. Research returns: work_orders/ccr/CREW/OUT/DOCKET/R1_*.md ... R4_*.md.

Not a patent application, not legal advice, not a novelty opinion. Filing is **Legal + Keith**. This TUI does not write v:\Ai Legal and does not click USPTO. **Do not post** whitepaper / X / LinkedIn / arXiv until provisionals are filed.

DOM rails still owed for official TESS + Patent Public Search (SGH + GEM). This pass used SSA + Google Patents + arXiv. TESS = UNKNOWN.

Clock (measured)

Repo	GitHub created_at	Notes
bts-mesh	2026-08-16T10:21:14Z	About text names “CRUCIBLE method.” Tracked files: 0 hits for CRUCIBLE.
cosmos	2026-08-23T06:42:12Z	Ratification. Architecture + MOTIF in the tree.
cdeck	2026-08-24T18:38:08Z	Product BUILD.

US: ~1 year inventor grace from *their* disclosure. Many other countries: absolute novelty - GitHub can already bar those filings for what it enabled. Private-now does not un-publish. Another public article dump is a second clock.

July 2026 operation = dated local predecessor records, **not** a GitHub date.

Fees

US provisional **micro-entity \$65** (37 CFR 1.16(d)); small \$130; large \$325. Provisionals do **not** count toward the micro-entity five-application cap. Trademarks are **\$350 per class**, not this stack. Keith certifies micro-entity.

Twelve packets

#	File	Title	Status
1	docket/P01_MOTIF_LOOP.md	8-stage; iterate to RESEARCH	FILE
2	docket/P02_DUAL_LANE.md	Dual-lane; no shared context	FILE
3	docket/P03_POROSITY.md	Family-axis; Porosity ; superlinear vs copies	FILE
4	docket/P04_RUNTIME_BIND.md	Live-emit gate vs green-log	FILE
5	docket/P05_ADVERSARIAL_ENGINE.md	Occupancy engine; one disposer; skins	FILE parent
6	docket/P06_LOCAL_FREE_WEIGHTS.md	Local free weights; fixed cost vs S-tier	FILE
7	docket/P07_CORE_LEDGER_FENCE.md	Sole signed JSONL + fence + tokens	FILE
8	docket/P08_DOM_FIRST.md	DOM-first unmetered rail	FILE

#	File	Title	Status
9	docket/P09_SPEND_GATE.md	Confirm-to-widen spend cap	FILE
10	docket/P10_RESOLVER.md	Sentinel root; existence $\neq$ identity	FILE narrow
11	docket/P11_SEED.md	Signed close; OPEN_CONTEXT	FILE narrow
12	docket/P12_CRUCIBLE.md	Crucible seats	HOLD (PDJ art granted in CN; US app claims the triad). 12th slot: P05-as-legal-skin or UPS-JUDGE with Keith's July pack.

Reliable = P02+P03+P04+P05. Scalable = P06 (local, free weights, low fixed cost). OS = P07-P11. Loop = P01.

Marks (knockout, not TESS)

COSMOS / MOTIF / Crucible / Forge / Docket = **crowded**. NVIDIA Cosmos is the practical COSMOS problem. **cDeck** and **Gitur** looked cleanest on secondary DBs. Confirm official Trademark Search before any \$350 filing.

Provenance (not a patent)

On-disk BTS-MESH (v:\Ai\BTS_MESH - read, do not write from this TUI): dated July run docs. Live ledger is gitignored. Do not commit keys or ledgers.

Vetting summary - Docket MOTIF 1 to 7 (2026-09-07)

Not a novelty opinion. Not a filing. Not TM clearance. Research returns: work_orders/ccr/CREW/OUT/DOCKET/R1_OCCUPANCY.md, R2_LOCAL_FREE.md, R3_CORE.md, R4_CRUCIBLE_TM.md. CCr dispose of packets P01-P12.

How MOTIF ran

Stage	What actually happened
1 RESEARCH	Four independent SSAs (occupancy, local free-weights, Core/fence, Crucible/TM). Google Patents + arXiv + products + X. FINDING: official USPTO TESS and SGH+GEM DOM rails not executed. Secondary TM DBs only.
2-3 ARCH/CONSENSUS	Packet shape _PACKET_SHAPE.md. No second full ARCH manuscript (FINDING: Copilot cannot write this tree). Research lanes disagreed on nothing material; crowding vs combination is the CONSENSUS.
4 BUILD	CCr wrote docs/research/docket/P01-P12 + this summary.
5-7 CRITICS/IMPROVE	CCr vs RESEARCH: Crucible PDJ HOLD ; Porosity named in P03; P06 must not claim MoA/FrugalGPT slogans; COSMOS/MOTIF/FORGE/CRUCIBLE marks crowded.

File vs HOLD

Packet	File?	Why
P01 MOTIF loop	FILE	Iterate-to-RESEARCH + dual-lane + vendor-plural not found as one named loop. Broad “staged MAS-SDLC” would die (ChatDev, CN121436020A).
P02 Dual-lane	FILE	Peeking ban + two full builders. OpenAI US12,405,822 B1 is anti-shape (shared workspace). Don’t claim “two agents in parallel.”
P03 Porosity	FILE	Named quantity + different-mistake overlay vs copies. Papers crowded (Hidden Clones, co-failure β). Patents thin . Do not claim Reason. Do not put an exponent in the spec.
P04 Runtime-bind	FILE	Thin patents. High crowding if “don’t trust CI.” Keep live-emit-only.
P05 Engine	FILE	Parent. Pieces crowded; combination UNKNOWN.
P06 Local free weights	FILE	MoA/FrugalGPT/Blending block slogans . Legal must read Microsoft US12524210B2 (local vs remote coding COGS) and Google US20240311405A1 (pick one of N models). Isolation (no shared context) is the thin element. No hit found that measures 2-3 isolated local free-weights vs one S-tier on box vs token dollars. Self-MoA: mixing can <i>hurt</i> .
P07 Core	FILE	US11943344B2 closest patent. Combination as AI OS UNKNOWN.
P08 DOM-first	FILE	Browser agents crowded. “Prefer because no credit” UNKNOWN as patented.
P09 Spend gate	FILE	Hard caps exist. Confirm-to-widen + signed 409 UNKNOWN as blocking.
P10 Resolver	FILE narrow	Empty-dir scar. Analog: OCI digest, Nix.
P11 SEED	FILE narrow	Watch CONTINUITY arXiv:2609.05269 (two days before this search).

Packet	File?	Why
P12 Crucible	HOLD	PDJ triad is granted in CN and claimed in US20260037351A1. File only as P05 skin. TM: CrucibleTech + Star Lab registered.

12th slot if you want 12 filings: UPS-JUDGE only with Keith's July pack (no external UPS-JUDGE product found). Or file P12 as "P05 applied to a legal packet" with **no** PDJ independent claims.

Marks (knockout, not TESS)

Name	Scan
COSMOS	Crowded Cl. 9/42; NVIDIA Cosmos is the practical AI collision.
MOTIF	Crowded ordinary English; software marks exist.
Crucible	Registered others (games, OS). Destiny fame.
Forge	Crowded (Forge Global, Ready Robotics, Shield AI, Forge AI).
Docket	Descriptive in legal; Clarivate product.
cDeck, Gitur	Cleanest on this incomplete scan. Confirm TESS.
KDash	US software open-looking; JP talent agency ■■■■■■.

Clock

Public GitHub already disclosed architecture + MOTIF (cosmos **2026-08-23**; bts-mesh About **2026-08-16**). File provisionals **before** posting whitepaper/X/LinkedIn/arXiv. US grace ~1 year. Absolute-novelty countries: already at risk for what GitHub enabled. Private-now does not un-publish.

Fees (USPTO)

Provisional micro-entity **\$65** (37 CFR 1.16(d)); small \$130; large \$325. Eleven FILE packets ~ **\$715** micro if all filed. Trademarks **\$350/class**, not this stack. Provisionals do not count toward micro-entity's five-application cap. Keith certifies micro-entity; CCr does not.

Porosity (this pass)

Keith: coder quality = **Porosity** (Pourosity); low-quality models with **different mistakes**, overlaid, cover surface **quickly**, > **linear vs copies**. Encoded P03. Superlinear is vs the same-family baseline (flat). Do not invent an exponent. Convert with a measurement: coverage(N) different-family vs same-family.

This TUI will not

Click USPTO. Write v: \A1 Legal. Post the four public pieces. Draft formal claims. Assert novelty.

Provisional packet shape (Legal, not USPTO)

Each Pxx_*.md is a **written-description outline** for a US provisional. Not a filed application. Not claims. Not a novelty opinion. Keith + Legal file.

Every packet has:

- 1 Working title + docket ID
- 2 Kind: method / system / occupancy
- 3 One-sentence what it is (in-tree)
- 4 Problem / scars that earned it
- 5 Written description (enough that Legal can draft a spec)
- 6 Already public (GitHub date + what leaked)
- 7 Prior art to name (from MOTIF 1 returns) - related vs possibly blocking = research judgment
- 8 What this packet is **not**
- 9 Suggested independent idea (one line, not a claim set)
- 10 Status: FILE / FOLD / HOLD / NEEDS KEITH

P01 - MOTIF eight-stage loop (iterate returns to RESEARCH)

Kind: method. **Status:** FILE. **Fee class:** US provisional, micro \$65 / small \$130 / large \$325.

What it is (in-tree)

An eight-stage development method for AI-built software: RESEARCH to ARCH to CONSENSUS to BUILD to CRITICS to CONSENSUS to IMPROVE to ITERATE. **ITERATE re-enters RESEARCH**, not critics. Encoded docs/MOTIF.md 2026-08-25. Vendor-plural required. Dual-lane BUILD and different-family CRITICS are seats of this loop (see P02, P03, P05).

Problem / scar

A critics-only loop (stages 5-8) can only polish the thing already built. A frozen coder and a frozen architecture stop evolving. New models, new facts, and shifted needs never re-enter.

Written description (outline for Legal)

- 1 Receive a work item that is not a one-line implement.
- 2 RESEARCH: returns on disk before BUILD; prefer rails that cannot run out of credit (see P08).
- 3 ARCH: rubric first; independent designs; no peeking.
- 4 CONSENSUS: converge or mark CONTESTED (both positions, one line to the human). No third model resolves.
- 5 BUILD: dual-lane, no shared context (P02). Each spike must run.
- 6 CRITICS: different family; judge against *what was decided*.
- 7 CONSENSUS: reconcile; prior versions are baselines.
- 8 IMPROVE: apply; subtract as well as add.
- 9 ITERATE: return to step 2 (RESEARCH). Re-decide architecture, primary coder, baselines.
- 10 Gate: runtime-binding (P04). Exit codes and green logs are not DONE.

Already public

keithbbf-gif/cosmos created **2026-08-23**. MOTIF.md in the tracked tree. US grace ~1 year from inventor disclosure. Absolute-novelty countries: already at risk for what GitHub enabled.

Prior art to name (R1)

Related / crowding: ChatDev arXiv:2307.07924 (waterfall; iterate stays in test); MetaGPT arXiv:2308.00352 (SOP + shared pool); AutoGen arXiv:2308.08155; AgentCoder arXiv:2312.13010; MapCoder arXiv:2405.11403 (debug reverts to *planning*, not research); Twilio **US20250165890A1** (planner-critic-executor); **CN121436020A** (eight-stage agent DevOps; Retrain to Prompt); **CN121029145B** (waterfall MAS); aiXplain **US20260099419A1** (refine-the-config loop); Boehm spiral 1988; Scrum inspect-and-adapt.

Not found: a US patent whose claims *require* the last stage to re-enter a *research* stage (external returns on disk) rather than test/critique. UNKNOWN unpublished apps.

What this is not

Not “any staged multi-agent SDLC.” Not ChatDev. Not a patent filing. Not a bake-off.

Suggested independent idea (not a claim set)

A development loop whose iterate step is required to re-enter research (fresh independent designs and new critics), rather than a critics-only polish cycle.

P02 - Dual-lane BUILD, no shared context

Kind: method / occupancy seating. **Status:** FILE. **Fee:** US provisional micro \$65.

What it is (in-tree)

Two builders on the same decided work, **no shared context** (no shared transcript, no peeking at the other lane's branch). Adversarial: two builders, **not** builder-plus-checker. Compare is a later consensus. Encoded MOTIF stage 4; docs/ADVERSARIAL_LOOP.md. Forge seats (`forge.ccr` + `forge.adv_N`) are a product seating.

Problem / scar

Shared-context debate lets a weak agent pollute a strong one (Wynn, Satija, Hadfield 2025). Builder-plus-nanny is the stacked slice: the second seat is hired to agree. OpenAI **US12,405,822 B1** (granted 2025) *teaches a shared workspace* - the anti-shape.

Written description

- 1 After CONSENSUS names the work, spawn **two** builder attempts.
- 2 Each attempt has a private workspace and a private context. Neither builder may read the other's context, branch, or intermediate artifacts until compare.
- 3 Each attempt must produce a running spike.
- 4 Compare is a later consensus step, not a silent merge by a third model.
- 5 The live tree has **one** writer (P05). Builders propose; they do not hold the pen.

Already public

Dual-lane described in public cosmos MOTIF (2026-08-23). Gitur dual-lane (Grok + Cursor) is operational occupancy, not a secret.

Prior art to name (R1)

Related: Galápagos N-version LLM arXiv:2408.09536; N-Version Programming with Coding Agents arXiv:2606.20158; Croto arXiv:2406.08979 (independent teams *then* interchange); IBM **US20260252812A1** (parallel contestants who later rank each other); Avizienis 1985; Knight & Leveson 1986. Twilio US20250165890A1 and AgentCoder = builder-checker (**anti-shape**). OpenAI US12,405,822 B1 shared workspace (**anti-shape**). Aider Architect/Editor: plan then edit - executor sees the plan.

Not found: a granted US patent that *forbids* two coding agents from reading each other's context as the inventive occupancy. UNKNOWN unpublished.

What this is not

Not "two agents code in parallel." Not N-version at runtime with a voter. Not shared-ledger multi-agent (OpenAI). Not builder-plus-linter.

Suggested independent idea

Isolating two full builders (each must run) with a peeking ban until compare, plus a single disposer on the live tree.

P03 - Family-axis occupancy; Porosity; superlinear cover

Kind: method / occupancy rule. **Status:** FILE. **Fee:** US provisional micro \$65.

Named quantity (Keith 2026-09-07): Porosity (written Pourosity in the order - use **Porosity** in the spec). The quality of a coder is quantified as porosity: the hole-set / mistake surface of that model (error classes, unseen tools, fabrication modes).

What it is (in-tree)

Same-vendor-family models are **one vote**, not two. Residual exposure is the **intersection** of hole-sets. Swiss-cheese image: Reason 1990 as *analogy*, not his accident-causation theory claimed as ours. Encoded docs/MOTIF.md 2026-09-07.

Keith 2026-09-07 (this pass): low-quality models with **DIFFERENT mistakes**, when **overlaid**, cover surface area **quickly**, at a **greater-than-linear** rate versus stacking copies. Copies of the same family leave holes aligned (coverage ~ flat). Complementary porosity overlays punch residual holes.

Hypothesis to **measure** (do not invent an exponent in the provisional): coverage vs N for different-family overlays vs same-family copies. Same-family incremental coverage ~ 0. Different-mistake overlay: residual ~ $p_1 \cdot p_2 \dots$ if holes are unaligned; cover of the remaining surface is large on the first different overlay. That is the “> linear vs copies / quickly cover” statement. Convert with a packet; do not put a fake number in the spec.

Problem / scar

N copies of one API (self-consistency; “More Agents Is All You Need”) bill N times and keep the same holes. Hidden Clones (2026) and LLMs-as-Jury (2026) already *measure* family-correlated errors. Knight & Leveson 1986: independently written versions do not fail independently.

Written description

- 1 Assign each model a **family** (training house + tool surface + observed failure mode). Occupancy rule, not a clustering theorem.
- 2 Quantify each coder’s **Porosity** $p(m)$ as a hole-set measure (error classes / residual miss surface). Exact metric is an embodiment (pass/fail per task class, β co-failure, etc.).
- 3 When seating votes (ARCH, BUILD, CRITICS): same family does not count twice.
- 4 Overlay: residual holes of the ensemble ~ intersection of porosities when mistakes **differ**; ~ the first porosity when they **align**.
- 5 Prefer adding a *different-porosity* (different-mistake) model, including a **low-quality** one, over adding a high-quality copy of the same family. Predicted: surface cover grows faster than the same-family baseline (flat / linear-in-invoice only).
- 6 Dual-lane + different-family critics exist to punch **different** holes (P02, P05).

Already public

Swiss-cheese occupancy in MOTIF.md on public cosmos (method text 2026-08-25; swiss cheese 2026-09-07 still in-tree). Family-axis papers 2026 are **their** measurements, not ours.

Prior art to name (R1)

Must cite: Reason 1990 / BMJ 2000 (metaphor); Knight & Leveson 1986; Hidden Clones arXiv:2603.17111 (family bias; 17 models to ~3 independent voters); LLMs-as-Jury arXiv:2607.10139; co-failure ceiling arXiv:2606.27288; Wisdom and Delusion of LLM Ensembles arXiv:2510.21513 (coding, 5 families); More Agents Is All You Need arXiv:2402.05120 (**anti-shape**: same-model copies); Self-consistency arXiv:2203.11171; Mixture-of-Agents arXiv:2406.04692; Self-MoA arXiv:2502.00674 (**adverse** to naïve mixing); Estornell & Liu NeurIPS 2024; DZone/Cognaptus “swiss cheese AI” as **safety-stack** metaphor (not occupancy).

Patents: thin. No US patent found that claims “same family is one vote” / porosity overlay. UNKNOWN unpublished.

What this is not

Not Reason’s aviation model. Not “ensembles exist.” Not a measured superlinear exponent. Not a clustering theorem.

Suggested independent idea

Quantified **Porosity** of each coder; seating and overlay rules so that **different-mistake** (including low-quality) models cover residual surface faster than same-family copies.

P04 - Runtime-binding gate vs green-log

Kind: method / selection gate. **Status:** FILE. **Fee:** US provisional micro \$65.

What it is (in-tree)

DONE is a **value only the live tree can emit**. Exit codes, critic prose, dashboard color, and this PDF are not evidence. Fail-closed: a missing or contradicting artifact is a refusal, not a silent pass. Encoded MOTIF gate; docs/SCAR_PLACATION.md (fabricated compliance).

Problem / scar

A pipeline reports success because tests exited zero or a critic said “looks good.” That is fabricated compliance - a plausible lie that closes the ticket while the running system does not do the thing.

Written description

- 1 For a change to be accepted, require an artifact that **only the running system** can produce (a live API value, a ledger event field, a hash only the true run emits).
- 2 Treat CI green, critic “LGTM,” and UI checks as **projections**, never authority.
- 3 If the live emit is missing or contradicts the claim, record a **refusal** (fail-closed). Do not repair a green log in place.
- 4 The report of the change must carry the live artifact or the refusal.

Already public

Runtime-binding language in public MOTIF / architecture (cosmos 2026-08-23).

Prior art to name (R1)

Related: Proof-or-Stop arXiv:2607.14890; Microsoft Azure 2026-08-29 “validate at runtime”; The New Stack runtime verification; CI as quality gate (Jenkins, GitHub Actions) - the *opposite* occupancy if CI is treated as DONE. No US patent found that claims “only a live-system-emitted value is DONE.” Patent occupancy **thin**. Paper/engineering occupancy **high 2026**.

What this is not

Not “don’t trust CI.” Not a test framework. Not a bake-off.

Suggested independent idea

Binding “finished” to a value only the live tree can emit, with fail-closed refusal when that value is absent.

P05 - Adversarial occupancy engine (the new way)

Kind: system / occupancy. **Status:** FILE (parent of products). **Fee:** US provisional micro \$65.

What it is (in-tree)

The engine: independent first, then argue; **no shared context** between builders; different-family critics vs *what was decided*; **one disposer** holds the pen; everyone else proposes. Products (Forge, Crucible, Diligence, Differentiator, Docket) are **skins**, not new inventions. MOTIF (P01) is the loop that *runs* this engine. Dual-lane (P02) is one seating. Keith: *the new way, reliable, scalable AI*.

Problem / scar

If the orchestrator also holds the only wrench, the system ships a plausible lie. Shared-transcript crews (AutoGen, ChatDev, MetaGPT, Magentic-One) put many agents in one context. Microsoft **US20250371498A1** teaches the second agent *based on* the first - anti-isolation.

Written description

- 1 Name seats: disposer (one writer), N proposers, critics.
- 2 Proposers run isolated; no shared transcript mid-pass.
- 3 Critics are a different family from the builders; they judge the decision, not house style.
- 4 Only the disposer writes the live tree (or the grant tree).
- 5 The same occupancy is reused across domains by changing the **packet** (code, casefile, deal room, IP docket), not by cloning the stack.

Already public

CRUCIBLE named in bts-mesh GitHub **description** 2026-08-16 (word not in tracked files - see R4). Adversarial loop in public cosmos MOTIF 2026-08-23.

Prior art to name (R3, R1)

Du 2023 debate; Irving 2018; Liang MAD; MoA; More Agents; Estornell & Liu; WO2025183627A1 (Lemon Inc MAD); US20240104125A1; US20250371498A1 (Microsoft, anti-isolation); AutoGen/ChatDev/MetaGPT/CAMEL/Magentic-One; Avizienis 1985; Knight & Leveson 1986; GitHub CODEOWNERS (VCS analog of one merger). **Combination** (isolation + family critics vs *decided* + one disposer + skins); UNKNOWN as blocking. Pieces crowded.

What this is not

Not “we invented multi-agent.” Not Magentic-One (orch holds the wrench). Not a voter over N-version binaries.

Suggested independent idea

One occupancy engine (isolate, different-family critique, one pen) reused as skins across coding/legal/medical/diligence/IP.

P06 - Local free-weight plurality vs S-tier (low fixed cost)

Kind: method / system of deployment. **Status:** FILE. **Fee:** US provisional micro \$65.

What it is (in-tree)

2-3 **free-weight** models of **different families, isolated, locally deployed** on the **same machine, low fixed cost** (hardware + power, not per-token), versus one **S-tier API**, on **price and task performance**. Keith: *locally deployable for a low fixed cost with free weight models*. Easy to show because some weights are free and several fit one box. **Not yet a controlled bake-off**. OpenRouter “free” APIs are a **different** rail (rotator refused). Combine with P03 (Porosity / different mistakes).

Problem / scar

S-tier is a variable invoice (quota, key, billing). One brain, same porosity. Metered “free APIs” still run out. Local free weights + isolation + family-axis is how cost becomes a **cap you can name**.

Written description

- 1 Select N=2 or 3 models whose **weights** are free/open and whose **families differ** (P03).
- 2 Deploy them **on-box** (llama.cpp / Ollama / vLLM class). Cost = machine + power.
- 3 Run them **isolated** (P02): no shared context mid-pass.
- 4 Dispose through one writer (P05). Gate on a live emit (P04).
- 5 Compare dollars and task outcome to a single S-tier API pass on the same task. Measurement is for conversion, not a number in this provisional.

Already public

Swiss-cheese price/performance *hypothesis* in MOTIF.md and the unposted publish pack. Local multiplex as Keith’s 2026-09-07 measurement plan - **not** a public bake-off.

Prior art to name (R2)

Must cite - blocks slogans, not necessarily the combination: Mixture-of-Agents arXiv:2406.04692 (open mix > GPT-4o AlpacaEval; **shares** layer context; cloud-scale open models); Blending Is All You Need arXiv:2401.02994 (2-3 small vs ChatGPT on **engagement**); FrugalGPT arXiv:2305.05176 (API **cascade**, up to 98% cost cut); RouteLLM arXiv:2406.18665; Hybrid LLM arXiv:2404.14618 (edge vs cloud, **one** model per query); More Agents arXiv:2402.05120 (same-model copies); Self-MoA arXiv:2502.00674 (**mixing can hurt**); LLM-Blender; PrivateGPT / GPT4All / LocalAI / OnPrem.LLM; llama.cpp --models-preset; Ollama multi-model; vLLM; Intelligence per Watt arXiv:2511.07885.

Patents Legal must read (R2 close): **US12524210B2** / WO2024238128A1 Microsoft (granted 2026-01-13) - hybrid **local GPT vs remote LLM** for coding COGS; **one** local + **one** remote, router predicts accept. **US20240311405A1** Google - pick **one** of N heterogeneous generative models (on-device small vs remote large). Citibank US12536406B2 gateway; Martian US12314825B2 prompt routing. On-device multi-LM **speculative decoding** (US12505335B2) **shares** the decode stream (anti-isolation).

Products: Ollama / llama.cpp multi-model = **tooling** (“run Ollama twice”). XDA 2026-08: Gemma + Qwen-Coder on one GPU, **user** switches tabs. AirgapAI: local multi-model, fixed license, **shared chat** (not E4).

Gap: this search found **no** paper/patent that measures **2-3 isolated local free-weights vs one S-tier API** on a coding task with **box dollars vs token dollars**. Combination E3+E4+E5+E6+E7: UNKNOWN as blocking (research, not a legal conclusion).

What this is not

Not “run Ollama twice.” Not FrugalGPT (paid API cascade). Not Together MoA (shared layers, hosted). Not a fake bake-off number.

Suggested independent idea

Family-diverse **free-weight** models, isolated, on one local machine at **fixed** cost, selected and gated as occupancy
- versus one metered S-tier pass.

P07 - COSMOS Core: sole ledger, fence, tokens

Kind: system. **Status:** FILE. **Fee:** US provisional micro \$65.

What it is (in-tree)

One resident service: sole writer of an **append-only, hash-chained, service-signed JSONL** ledger. Dashboards/SQLite = **rebuildable projections**, never authority. **Leases + monotonic fencing tokens**. Workers publish only through a **fenced commit gateway** (token + expected input hashes). Corrupt segment **REFUSES**, never repair-in-place. Ratified docs/FINAL_ARCHITECTURE.md 2026-08-23.

Problem / scar

Two writers, empty-dir identity, advisory locks, “repair the log.” Green dashboards as authority.

Written description

- 1 One process is API + scheduler + arbiter + ledger writer.
- 2 Ledger: framed JSONL, hash chain, service signature. Replay is the rebuild of projections.
- 3 Projections (queue, KDash, spend totals) may not be written as authority.
- 4 Workers get a fencing token; commit gateway rejects stale tokens and hash mismatches.
- 5 Fail-closed on corrupt segment.

Already public

Architecture text on cosmos **2026-08-23**. Predecessor mesh named on bts-mesh **2026-08-16**.

Prior art to name (R3)

Event sourcing (Fowler 2005); CQRS; git; Bitcoin; Kafka; **US11943344B2** Ridgeline (granted 2024 - hash-chained signed events + projections, closest patent); **WO2018217375A1** Microsoft signed log-chain; RFC 6962 Certificate Transparency; Kleppmann fencing tokens 2016; Chubby 2006; Raft 2014; etcd/k8s leases; SCSI-3 PR / STONITH; Temporal; Airflow; n8n; LangGraph. **Combination of all five as an AI-work OS:** UNKNOWN as blocking.

What this is not

Not blockchain. Not git. Not Temporal. Not “event sourcing exists.”

Suggested independent idea

An AI-work OS whose authority is a single-writer signed hash-chained JSONL plus fencing-token commit of worker artifacts, with projections never authority.

P08 - DOM-first as first-class unmetered rail

Kind: method / routing policy. **Status:** FILE. **Fee:** US provisional micro \$65.

What it is (in-tree)

The **browser/DOM** is a first-class scheduler rail and the **preferred** rail because it depends on nothing that can run out (credit, quota, billing, key, consent). Metered APIs are explicit, audited fallback. Contained DOM workers: own OS identity, ephemeral profiles, Job Objects, typed failures UNREACHABLE / SESSION_EXPIRED / AUTH_REQUIRED / BROKE. AD-6.

Problem / scar

API-only agents die when the key, quota, or vendor dies. DOM is usually a *tool*, not the preferred unmetered path.

Written description

- 1 Register DOM workers as a scheduler rail, not a one-off script.
- 2 Routing policy data: DOM first, API second; fallback is explicit and ledgery.
- 3 Typed failures; report-never-retry unless contract-idempotent.
- 4 Prefer this rail for RESEARCH (P01) because returns must not depend on prepaid remaining.

Already public

DOM-first in public architecture 2026-08-23.

Prior art to name (R3)

Playwright; Playwright MCP; browser-use; Stagehand; Skyvern; Magentic-One WebSurfer; **US12101373B2** (browser RPA); **CN119248379B** (LLM + Playwright scheduler); computer-use / Operator-class **metered** products. Distinctive “prefer DOM **because it cannot run out of credit**” as routing policy: **UNKNOWN** as patented. Ops folklore is widespread.

What this is not

Not “browser automation exists.” Not WICG scheduler.postTask. Not OpenAI Operator (metered).

Suggested independent idea

Scheduler routing that prefers a contained DOM rail *because* it cannot exhaust credit, with typed failures and audited API fallback.

P09 - Spend gate fail-closed; confirm-to-widen

Kind: method / system. **Status:** FILE. **Fee:** US provisional micro \$65.

What it is (in-tree)

A spend cap **cannot silently widen**. POST without explicit confirm to **HTTP 409** WIDEN_REQUIRES_CONFIRM. Cap does not move on the 409. Signed BUDGET_SET / SPEND_CAP_REFUSED on the ledger. Measured docs/CHANGELOG_2026-08-30_CC_AUDIT.md.

Problem / scar

OpenAI removed hard budget limits (notify-only). Admin APIs that raise max_budget without a two-step confirm. Silent fallback reroute that spends a different rail.

Written description

- 1 Each rail has a signed cap.
- 2 A request that would raise the cap without a confirm token is refused (typed 409). The stored cap is unchanged.
- 3 Widen requires an explicit confirm on the same cap object; ledger the grant or the refusal.
- 4 Fail-closed: over-cap work does not run.

Already public

Architecture / changelog in public cosmos tree (2026-08-23+).

Prior art to name (R3)

LiteLLM virtual keys / tag budgets (admin can raise); Cloudflare AI Gateway spend limits 2026-06 (block by default; Dynamic Route can reroute); Anthropic workspace limits; LangSmith evaluator weekly cap; **US20250299128A1** multi-cloud budget throttle; EP4381451A1; PCI dual-control; 21 CFR 11.10. OpenAI hard-cap **removal** is the scar, not a teaching of confirm-to-widen. Combination (AI-rail cap + no silent widen + typed confirm + signed ledger): **UNKNOWN as blocking**.

What this is not

Not AWS Budgets alerts. Not “circuit breaker exists.”

Suggested independent idea

An AI-rail spend cap that cannot move without a typed confirm, with the refusal itself ledgery.

P10 - Resolver: sentinel root; existence is not identity

Kind: system (narrow). **Status:** FILE (narrow). **Fee:** US provisional micro \$65.

What it is (in-tree)

Runtime root is verified by **sentinel content** (.cosmos-root.json system + tree_id) + install record. Service cannot go READY without it. **No** drive literal as identity, **no** parent-walk, **no** fallback ladder. An empty directory that *exists* is not the tree (empty-dir scar). AD-5.

Problem / scar

Empty-dir: a path that exists is treated as the install. Fallback ladders silently pick the wrong tree. Two writers on two “roots.”

Written description

- 1 At boot, resolve one root by reading a sentinel **file content**, not by path existence.
- 2 Match declared system/tree_id; refuse READY on mismatch or missing install record.
- 3 Do not walk parents, do not try a list of drives, do not treat an empty dir as identity.

Already public

Resolver rules in public architecture 2026-08-23.

Prior art to name (R3)

OCI digest vs tag; Nix require-sigs / sandbox-fallback (fallback is the anti-pattern); git objects; chroot (path, not content); systemd-nspawn --root-hash; SLSA/in-toto/Cosign; HSTS fail-closed. **No close patent** found on JSON sentinel + refuse empty-dir + no parent-walk as AI-OS boot identity. UNKNOWN as blocking. Narrow.

What this is not

Not chroot. Not Docker tags. Not a new hash algorithm.

Suggested independent idea

Boot identity of an AI runtime root by sentinel *content*, where existence of a directory is not identity and fallback ladders are forbidden.

P11 - SEED / signed session close (OPEN_CONTEXT)

Kind: method / system (narrow). **Status:** FILE (narrow). **Fee:** US provisional micro \$65.

What it is (in-tree)

Close of a session writes a **signed** context manifest (SEED.json + declared length/HMAC). Inherited facts, active leases, open watchers, handoff recipient. Close **without** a valid manifest $\Rightarrow$ **OPEN_CONTEXT** incident. Start refuses NO_SEED / BAD_SEED / IDENTITY_MISMATCH. AD-10. Not ChatGPT “memory.”

Problem / scar

Sessions vanish. The next session fabricates continuity. Memory features store facts without a fail-closed close.

Written description

- 1 close_session must emit a signed manifest covering declared fields.
- 2 Absence, length mismatch, or HMAC failure is an **incident**, not a warning.
- 3 start_session reads the manifest under its declared length/HMAC before injecting carry-over.
- 4 Optional: auto-resession clock that consumes the same SEED (embodiment).

Already public

AD-10 in public architecture 2026-08-23.

Prior art to name (R3)

CONTINUITY arXiv:2609.05269 (2026-09-05 - signed context manifests + CLOSE/OPEN receipts; **closest paper; watch**); Portable Agent Memory arXiv:2605.11032; Context Passport; CRAFT handoff; ctx handover; vendor memory; CRIU; JWT/JWS; **US20250259069A1** reconstitutable sessions. Incident-on-missing-close as OS rule: **UNKNOWN as patented**.

What this is not

Not ChatGPT memory. Not CRIU. Not “write a notes file.”

Suggested independent idea

Session identity that *must* close with a signed manifest; missing close is an incident and start refuses.

P12 - Crucible (legal seats) - HOLD unless occupancy, not PDJ

Kind: method / application of P05. **Status:** **HOLD** as a 12th provisional unless Legal files occupancy-not-roles. **Fee:** \$65 only if filed.

What it is (in-tree)

Named multi-model legal round: independent then argue; plaintiff / defense / judge seats; vendor-plural + spend-gated critics. Public bts-mesh **GitHub description** (2026-08-16) says “CRUCIBLE method.” Tracked files in that repo: **zero** hits for the word CRUCIBLE (R4). So the public date is the **About text**, not a method disclosure in code.

Problem / scar

Need a legal Crucible as a *skin* of P05, not a clone of SimuCourt.

Written description (only if filing)

Do **not** claim “AI plaintiff, defense, and judge.” That triad is crowded (SimuCourt/AgentsCourt 2024, AgentCourt 2024, CN119168059B **granted** 2025-07-15, US20260037351A1 claim 21). File only if the spec is: **P05 occupancy** applied to a legal packet (isolate, different-family, one disposer, runtime-bind, ledger). Roles are embodiments.

Already public

Name “CRUCIBLE method” on GitHub About **2026-08-16**. US grace for what that string disclosed is thin (a name, not a spec).

Prior art to name (R4) - blocking for role-triad claims

SimuCourt/AgentsCourt arXiv:2403.02959; AgentCourt arXiv:2408.08089; LegalSim; SAMVAD; PROClaim arXiv:2603.28488; 3-Ply arXiv:2606.30906; **CN119168059B granted; US20260037351A1; US20250148558A1** (jury-side). Products: Jury Simulator, Juries.ai, Trial AI.

Name collisions: CrucibleTech RN **8043152** (games); Star Lab CRUCIBLE RN **5023199** (OS); Crucible Discovery (eDiscovery); Destiny PvP fame.

What this is not

Not “we invented mock trial AI.” Not a TM clearance.

Suggested independent idea

If filed: Crucible as a **skin of P05**, not a PDJ patent. Else fold into P05 and spend the 12th slot on **UPS-JUDGE** only with Keith’s July pack (R4: no external UPS-JUDGE product found; do not invent the spec).